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Keyword Networks, Thematic Evolution and Growth Forecasts of Agricultural Drone Research to 2035

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18,506 articles, 2005-2025, for growth models

28.5 % annual growth (about 32 % precision-adjusted)

5 stable research themes (Q = 0.41)

~21,000 articles forecast for 2035

~11,000 conservative lower bound for 2035

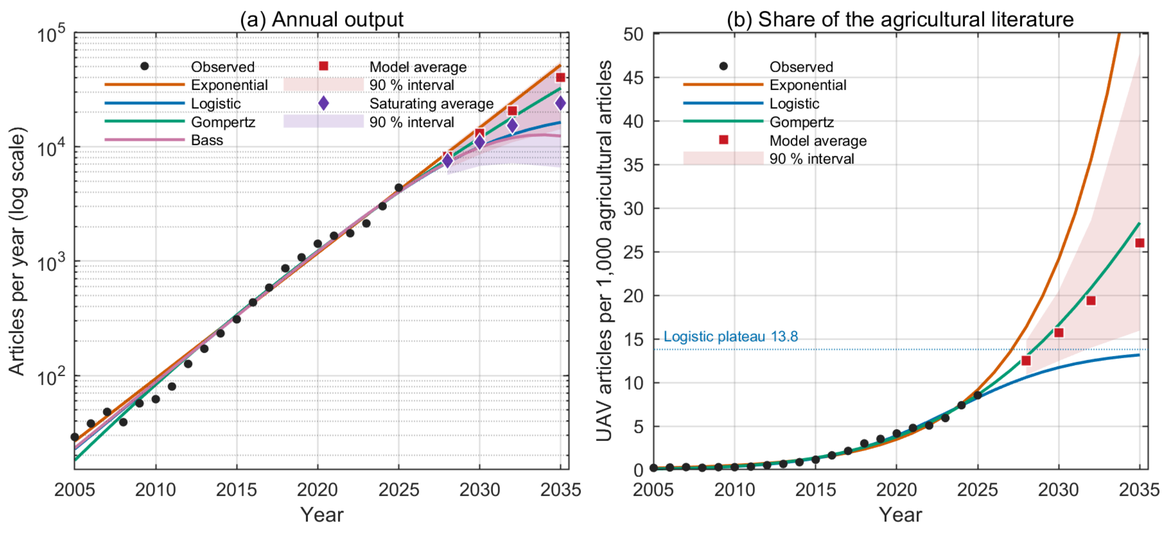
1. Aim

How fast is research on agricultural drones growing, which themes drive it, and when will it saturate? The study combines keyword science mapping with growth forecasting on one open corpus.

2. Data and methods

- Corpus A: 18,506 OpenAlex articles (2005-2025) combining drone and agricultural terms, for the growth models
- Corpus B: 22,357 records with keywords (2005-2026) for science mapping, after removing 723 homonym records
- Network of the 60 most frequent keywords, links of at least 50 co-occurrences, Louvain clusters stable over 50 seeds
- Thematic evolution across five periods and a heatmap of emerging keywords
- Four growth models with QAICc weights, backtests and bootstrap; long-horizon forecast = share x agricultural literature (+4.7 % per year)

3. Growth of the field



Output grew from 29 articles in 2005 to 4,381 in 2025. Corrected for the lower query precision of the early years, growth is about 32 % per year. The share of the agricultural literature is levelling off near its inflection point.

4. Forecasts to 2035

Approach	2028	2035
Share (model average) x literature	**7,400**	**21,200 (13,000-38,800)**
Share (logistic) x literature	6,200	10,700
Saturating count models	7,500	24,000
Exponential count model (rejected)	8,800	51,800

Counts are reliable about three years ahead. In the long run the share levels off, so output follows the agricultural literature; the share data reject the exponential path.

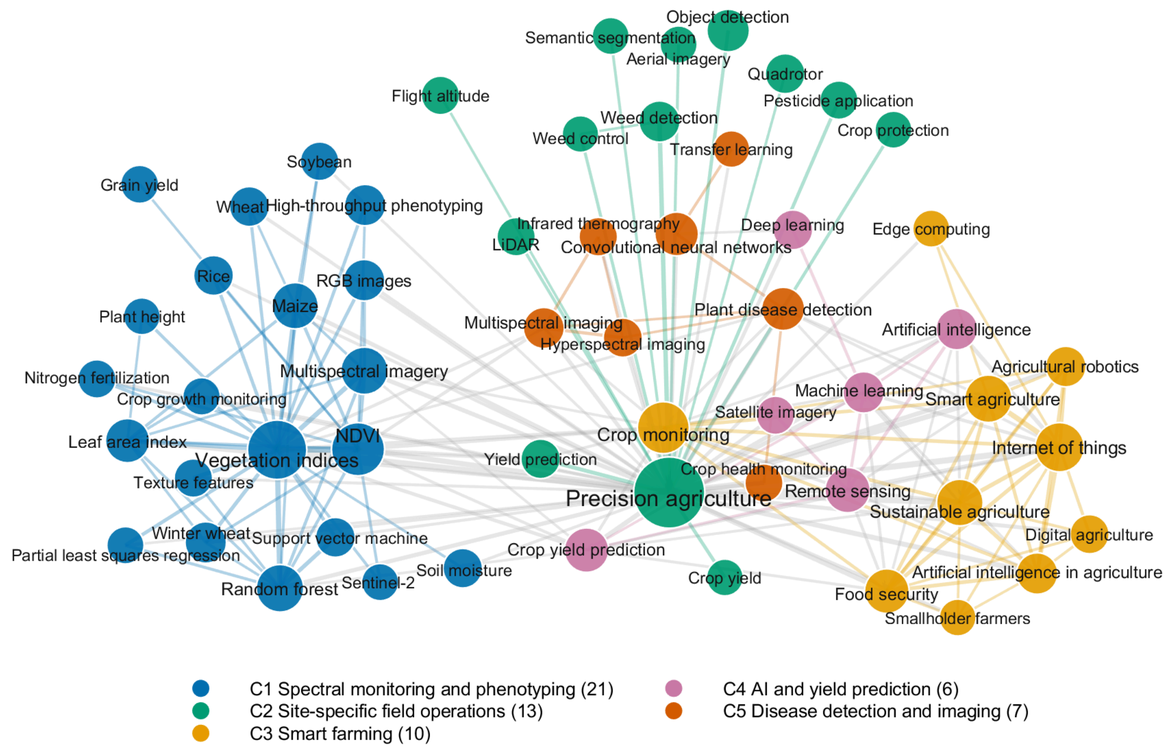
Implications

- Technology providers: the advantage shifts from aircraft hardware to analytics and on-board processing.
- Farms and cooperatives: shared ownership and drone services answer the cost barrier on small farms.
- EU policy: disease and irrigation monitoring can grow under current rules, while spraying needs a derogation.

Conclusion

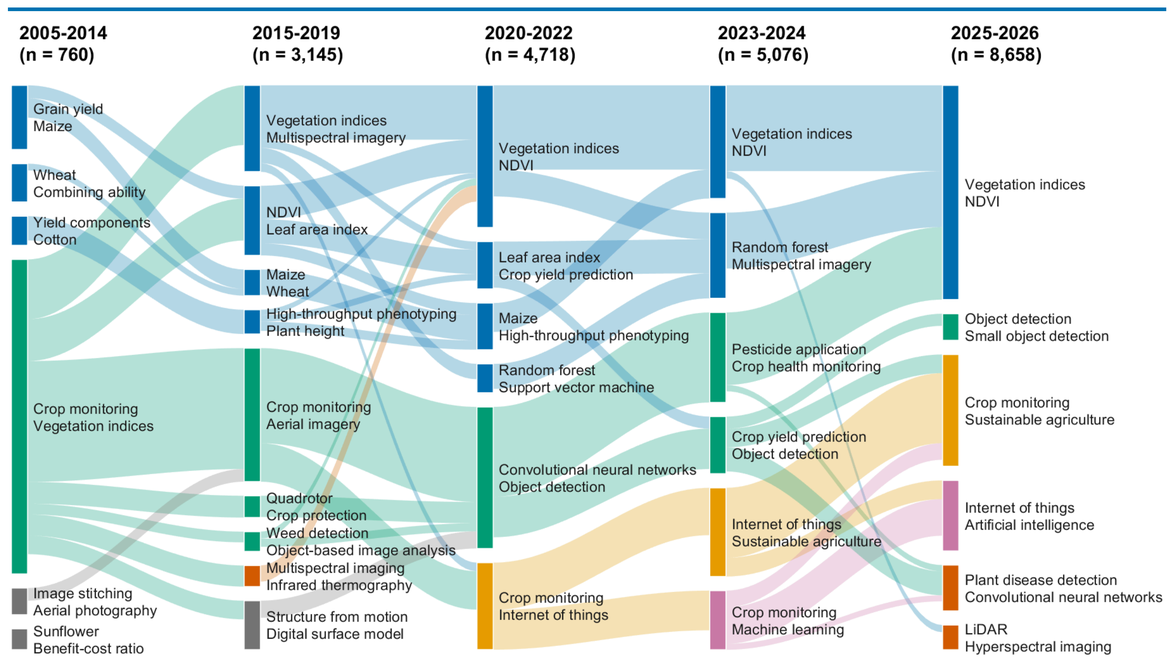
Agricultural drone research keeps growing, but it is near its inflection point. Its focus is moving from vegetation indices to deep learning, disease detection and smart farming. Output is likely to reach about 21,000 articles in 2035, with a lower bound of about 11,000.

5. Five research themes



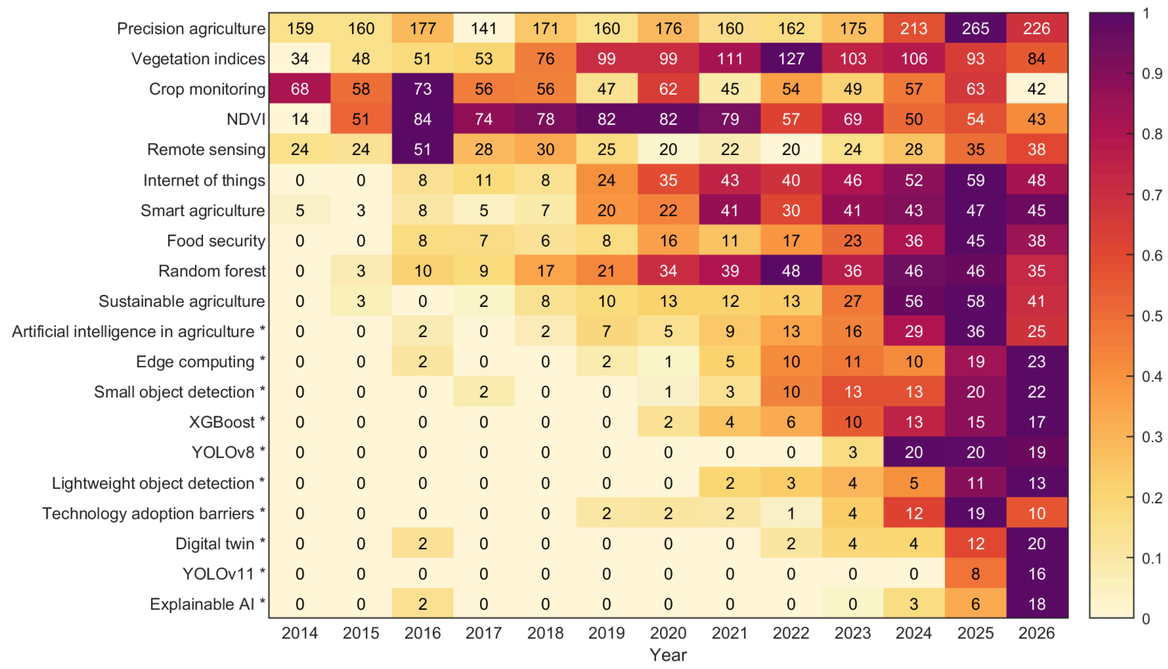
Spectral monitoring and smart farming form two poles, with precision agriculture as the hub.

6. Thematic evolution, 2005-2026



One broad monitoring theme split into deep learning, disease detection, smart farming and LiDAR.

7. Emerging keywords



YOLOv8 and YOLOv11, edge computing, digital twins and explainable AI rise fastest since 2023.

Scan for the data, code and figures

github.com/vedranu/drone-agriculture-forecast

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